

Training Agenda
IAEA Regional Training Course on
Development of New Mutant Germplasm and Populations of
all Target Food Crops with High Nutrient Content
Bangkok, Thailand
11 to 15 May 2026

Organizers: International Atomic Energy Agency (IAEA) in collaboration with the Government of Thailand through the Thailand Institute of Nuclear Technology (TINT)

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Venue: Avani Ratchada Bangkok Hotel, Sapphire 1 Room, 2nd floor
1 Ratchadaphisek Rd, Din Daeng, Bangkok 10400 Thailand
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Day 1: Monday, May 11, 2026

Time	Topic	Speaker
8:30 - 9:00	Registration	Host Organization
09:00 – 09:15	Opening Remarks	Host Organization (IAEA and TINT representative)
09:15 – 09:30	Overview of Training Objectives and Expected Outcomes	Host Organization
09:30 – 10:00	Participants' Introduction and Expectation Sharing	All Participants
10:00 – 10:30	Pre Test	Host Organization
10:30 – 10:45	Group Photo and Coffee Break	
10:45 – 12:00	Lecture 1: <i>Biofortification through Mutation Breeding: Rationale and Regional Priorities</i>	Dr. Xianmin Diao
12:00 – 13:00	Lunch Break	
13:00 – 14:30	Lecture 2: <i>Prioritizing Nutritional Targets and Defining Nutrition-focused Breeding Objectives in Food Crops</i>	Dr. Xianmin Diao
14:30 – 14:45	Coffee Break	
14:45 – 16:15	Lecture 3: <i>Physical and Chemical Mutagenesis Approaches in Crop Improvement</i>	Dr. Xianmin Diao

Day 2: Tuesday, May 12, 2026

Time	Topic	Speaker
09:00 – 09:45	Lecture 4: <i>Dose Optimization and Treatment Protocol Design</i>	Dr. Xianmin Diao
09:45 – 10:30	Practical Session 1: <i>Practical Exercise on Dose Optimization and Treatment Protocol Design</i> Note: Participants are requested to bring their own laptops.	Dr. Xianmin Diao and Local team
10:30 – 10:45	Coffee Break	
10:45 – 11:45	Lecture 5: <i>Nutritional Trait Analysis in Biofortification Breeding</i>	Dr. Xianmin Diao
11:45 – 12:45	Lunch Break	
12:45 – 13:30	Travel to Kasetsart University (KU), Bangkhen campus	
13:30 – 14:30	Site visit 1: <i>Faculty of Agriculture, Department of Agronomy</i>	Asst. Prof. Dr. Weerachai Matthayathaworn, KU
14:30 – 14:45	Coffee Break	
14:45 – 16:30	Site visit 2: <i>TINT, Bangkok Research reactor visit</i>	Local Speaker (TINT)
17:00 – 19:00	Dinner reception	
19:00 – 20:00	Return to hotel	

Day 3: Wednesday, May 13, 2026

Time	Topic	Speaker
9:00 - 10:15	Lecture 6: <i>Molecular Genetics and Tools in Mutation Breeding for Biofortification: TILLING and Beyond</i>	Dr. Kanokporn Boonsirichai, TINT
10:15 – 10:30	Coffee Break	
10:30 – 12:00	Lecture 7: <i>Integrating Molecular and Phenotypic Selection in Mutation Breeding</i>	Dr. Xianmin Diao
12:00 – 13:00	Lunch Break	
13:00 – 13:45	Lecture 8: <i>Case Study on Biofortification in Horticultural Crops</i>	Asst. Prof. Dr. Tassanai Jaruwattanaphan, KU
13:45 – 14:15	Lecture 9: <i>Case Study Biofortification through Mutation Breeding in Thailand: Rice (high Fe and Zn)</i>	Ms. Piyanuch Orpong, TINT
14:15 – 14:45	Practical Session 2: <i>Mutant screening using automation script (continuation of Lecture 10: Thailand Rice case study)</i> Note: Participants are requested to bring their own laptops.	Dr. Roppon Picha, TINT
14:45 – 15:00	Coffee Break	
15:00 – 15:45	Lecture 10: <i>Case Study Biofortification through Mutation Breeding in Thailand: Rice (high antioxidant)</i>	Asst. Prof. Dr. Weerachai Matthayathaworn, KU
15:45 – 16:30	Lecture 11: <i>Case Study on Biofortification in Tuber plant</i>	Assoc. Prof. Dr. Pitipong Thobunluepop, KU

Day 4: Thursday, May 14, 2026

Time	Topic	Speaker
8:00 - 9:00	Travel to Pathum Thani Rice Research Center, Pathum Thani	
9:00 - 11:00	Site Visit 3: <i>Rice Field and Laboratory visit, Pathum Thani Rice Research Center, Pathum Thani</i>	Ms. Ruenrudee Kaewchuenchai, Pathum Thani RRC
11:00 - 11:45	Travel to TINT Headquarters, Nakhon Nayok	
11:45 - 12:45	Lunch Break	
12:45 - 13:00	Site Visit 4: <i>TINT, Nakhon Nayok</i> Introduction to irradiation facilities and safety briefing	Dr. Roppon Picha, TINT
13:00 - 13:30	WD-XRF elemental composition analysis lab visit	Mr. Ratchai Funklin and Ms. Piyanuch Orpong, TINT
13:30 - 14:00	Plant Mutation Breeding Lab visit	Local Speaker (TINT)
14:00 - 14:15	Gamma chamber visit	Local Speaker (TINT)
14:15 - 14:30	Coffee Break	
14:30 - 15:30	Gamma room and E-beam visit	Dr. Apichet Maneewong, TINT
15:30 - 17:00	Return to hotel	

Day 5: Friday, May 15, 2026

Time	Topic	Speaker
09:00 – 10:15	Lecture 12: <i>Integrated Management and Selection of Mutant Lines in Breeding Pipelines</i>	Dr. Xianmin Diao
10:15 – 10:30	Coffee Break	
10:30 – 12:00	Lecture 13: <i>Colored Rice Varieties: A Turning Point in Nutritional Food Systems</i>	Prof. Dr. Apichart Vanavichit, Rice Gene Discovery / Rice Science Center
12:00 – 13:00	Lunch Break	
13:00 – 14:00	Summary of training (Q&A)	Dr. Xianmin Diao
14:00 – 14:30	Post Test, Course Evaluation and Feedback (2-4 representative participants)	Host Organization
14:30 – 15:00	Closing Remarks and Certificate Distribution	Host Organization
15:00 – 15:15	Coffee Break	

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